

Section A

Question				Marking details	Marks available														
					AO1	AO2	AO3	Total	Maths	Prac									
1				C ₉ H ₁₈ O ₆		1		1											
2				butanoic acid accept methylpropanoic acid / pentanedioic acid	1			1		1									
3				<table><tr><td>Proton(s)</td><td>Splitting pattern</td><td>Relative peak area ratio</td></tr><tr><td><i>a</i></td><td>singlet</td><td>1</td></tr><tr><td><i>b</i></td><td>singlet</td><td>2</td></tr></table>	Proton(s)	Splitting pattern	Relative peak area ratio	<i>a</i>	singlet	1	<i>b</i>	singlet	2			2	2		
Proton(s)	Splitting pattern	Relative peak area ratio																	
<i>a</i>	singlet	1																	
<i>b</i>	singlet	2																	
4				least acidic B C A most acidic		1		1											
5				C - (despite the lack of a solvent front measurement) the spots will ascend in order of their <i>R_f</i> values	1			1											
6				NaOH / I ₂ or NaOCl / KI pentan-2-one gives a yellow solid / precipitate and no observable change with 1-phenylbutanone		1 1		2		2									